

NLA Note

1 Background

Floating Point ($\mathbb{F}$): A number = mantissa $\times$ base^{exponent} (e.g. $2048 = 1 \times 2^{11}$ mantissa=1, base=2, exponent=11)

Assumptions: There is $\varepsilon_m > 0$ (**machine epsilon**) **A1:** $\forall \alpha \in \mathbb{R}, \exists \varepsilon \in (-\varepsilon_m, \varepsilon_m)$ s.t. $fl : \mathbb{R} \rightarrow \mathbb{F}$ by $fl(\alpha) = \alpha(1 + \varepsilon)$

$$\varepsilon_m \approx 2.22 \times 10^{-16}$$

A2: $\forall \alpha, \beta \in \mathbb{F}, \forall * \in \{+, -, \times, \div\}, \exists \varepsilon \in (-\varepsilon_m, \varepsilon_m)$ s.t. $\alpha * \beta = (\alpha * \beta)(1 + \varepsilon)$

A1: The rounding error in α is relative to the size of α ($|\alpha - fl(\alpha)| < |\alpha|\varepsilon_m$) **A2:** The error in $\alpha * \beta$ is relative to the size of $\alpha * \beta$ ($|\alpha * \beta - fl(\alpha * \beta)| \leq |\alpha * \beta|\varepsilon_m$)

Cost of Algorithm (C): $C :=$ number of floating point operations (FLOPs) If $C(n) \leq \alpha f(n), \Rightarrow C(n) = \mathcal{O}(f(n))$.

· We consider that: 交换/赋值/相反数 no cost (i.e. Permutation **P** is free), 比较两数 1 FLOPs, $\sqrt{\quad}$ 1 FLOPs.

Norms: $\|\mathbf{x}\|_p := (\sum_{i=1}^n |x_i|^p)^{1/p}, p \geq 1$ $\|\mathbf{x}\|_\infty := \max_{1 \leq i \leq n} |x_i|$ $\|\mathbf{x}\|_2 := (\sum_{i=1}^n |x_i|^2)^{1/2}$ $\|\mathbf{x}\|_1 := \sum_{i=1}^n |x_i|$

$$1. \|\mathbf{x}\|_p \geq 0 \quad \|\mathbf{x}\|_p = 0 \Leftrightarrow \mathbf{x} = \mathbf{0}$$

$$2. \|\alpha \mathbf{x}\|_p = |\alpha| \|\mathbf{x}\|_p \text{ for } \alpha \in \mathbb{R}$$

$$3. \|\mathbf{x} + \mathbf{y}\|_p \leq \|\mathbf{x}\|_p + \|\mathbf{y}\|_p \quad (\text{Triangle inequality})$$

$$4. \text{Induced Norm: } \|\mathbf{A}\|_p := \sup_{\mathbf{x} \neq \mathbf{0}} \frac{\|\mathbf{A}\mathbf{x}\|_p}{\|\mathbf{x}\|_p} = \sup_{\|\mathbf{x}\|_p=1} \|\mathbf{A}\mathbf{x}\|_p$$

$$1. \|\mathbf{A}_{m \times n}\|_\infty := \max_{1 \leq j \leq m} \sum_{i=1}^n |a_{ij}| \quad (\text{maximum row sum})$$

$$2. \|\mathbf{A}_{m \times n}\|_1 := \max_{1 \leq i \leq n} \sum_{j=1}^m |a_{ij}| \quad (\text{maximum column sum})$$

$$3. \|\mathbf{A}_{m \times n}\|_2 := \sqrt{\rho(\mathbf{A}^\top \mathbf{A})} \quad \rho(\mathbf{A}^\top \mathbf{A}) = \max\{|\lambda| : \lambda \text{ is eigenvalue of } \mathbf{A}^\top \mathbf{A}\}$$

$$\text{For Matrix: } \|\mathbf{A}\mathbf{x}\|_p \leq \|\mathbf{A}\|_p \|\mathbf{x}\|_p \quad \|\mathbf{AB}\|_p \leq \|\mathbf{A}\|_p \|\mathbf{B}\|_p \quad \|\mathbf{A}^k\|_p \leq \|\mathbf{A}\|_p^k \quad \|\mathbf{A} + \mathbf{B}\|_p \leq \|\mathbf{A}\|_p + \|\mathbf{B}\|_p$$

2 Methods for Systems of Linear Equations (SLE)

Ideal: We want to solve SLE: $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A}_{n \times n}$ is invertible, $\mathbf{x}, \mathbf{b} \in \mathbb{R}^n$

2.1 Backward Stable | Condition Number | Error

Backward Stable: An algorithm to solve SLE is Backward Stable if: $\exists \alpha(n) > 0$ s.t. $\forall \mathbf{A}, \mathbf{b}$, the computed solution $\hat{\mathbf{x}}$ satisfies:

$$(\mathbf{A} + \Delta \mathbf{A})\hat{\mathbf{x}} = \mathbf{b} + \Delta \mathbf{b} \quad \text{and} \quad \frac{\|\Delta \mathbf{A}\|_p}{\|\mathbf{A}\|_p} + \frac{\|\Delta \mathbf{b}\|_p}{\|\mathbf{b}\|_p} \leq \alpha \varepsilon_m \quad \varepsilon_m : \text{machine epsilon.} \quad \text{ps: } \alpha \text{ 可以和问题规模 } n \text{ 相关, 但不能和 } \mathbf{A} \text{ 或 } \mathbf{b} \text{ 相关.}$$

Condition Number ($\kappa_p(\mathbf{A})$): $\kappa_p(\mathbf{A}) := \begin{cases} \frac{\|\mathbf{A}\|_p \|\mathbf{A}^{-1}\|_p}{1} \geq 1 & \text{if } \mathbf{A} \text{ is invertible} \\ \infty & \text{otherwise} \end{cases}$ $\begin{cases} \text{If } \kappa_p(\mathbf{A}) \text{ is large (e.g. } \kappa_p(\mathbf{A}) \geq 10^{12} \approx \varepsilon_m^{-1}), & \mathbf{A} \text{ is ill-conditioned.} \\ \text{If } \kappa_p(\mathbf{A}) \leq 10^4, & \mathbf{A} \text{ is well-conditioned.} \end{cases}$ ps: $\kappa_p(\mathbf{A})$ 衡量矩阵奇异性

1. For $\mathbf{A}_{n \times n}$ with real eigenvalues $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$ and corresponding eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$: ps: 对所有实对称矩阵都满足下面条件

$$\text{If } \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n \text{ are orthonormal set. Then } \kappa_2(\mathbf{A}) = \|\mathbf{A}\|_2 \|\mathbf{A}^{-1}\|_2 = \frac{|\lambda_1|}{|\lambda_n|} \geq 1$$

Relative Error: $\frac{\|\hat{\mathbf{x}} - \mathbf{x}\|_p}{\|\mathbf{x}\|_p}$ **Actual Error:** $\|\hat{\mathbf{x}} - \mathbf{x}\|_p$ ps: Relative error gives more information than actual error.

1. Let $\mathbf{Ax} = \mathbf{b}$ and $(\mathbf{A} + \Delta \mathbf{A})\hat{\mathbf{x}} = \mathbf{b}$. If $\kappa_p(\mathbf{A}) \frac{\|\Delta \mathbf{A}\|_p}{\|\mathbf{A}\|_p} < 1$ then $\frac{\|\mathbf{x} - \hat{\mathbf{x}}\|_p}{\|\mathbf{x}\|_p} \leq \frac{\kappa_p(\mathbf{A})}{1 - \kappa_p(\mathbf{A}) \frac{\|\Delta \mathbf{A}\|_p}{\|\mathbf{A}\|_p}} \cdot \frac{\|\Delta \mathbf{A}\|_p}{\|\mathbf{A}\|_p}$. ps: $\frac{\|\Delta \mathbf{A}\|_p}{\|\mathbf{A}\|_p} \frac{\|\Delta \mathbf{b}\|_p}{\|\mathbf{b}\|_p}$ 小代表 good stability.

2. Let $\mathbf{Ax} = \mathbf{b}$ and $(\mathbf{A} + \Delta \mathbf{A})\hat{\mathbf{x}} = \mathbf{b} + \Delta \mathbf{b}$. If $\kappa_p(\mathbf{A}) \frac{\|\Delta \mathbf{A}\|_p}{\|\mathbf{A}\|_p} < 1$ then $\frac{\|\mathbf{x} - \hat{\mathbf{x}}\|_p}{\|\mathbf{x}\|_p} \leq \frac{\kappa_p(\mathbf{A})}{1 - \kappa_p(\mathbf{A}) \frac{\|\Delta \mathbf{A}\|_p}{\|\mathbf{A}\|_p}} \cdot (\frac{\|\Delta \mathbf{A}\|_p}{\|\mathbf{A}\|_p} + \frac{\|\Delta \mathbf{b}\|_p}{\|\mathbf{b}\|_p})$.

2.2 LU Factorization | Gaussian Elimination (GE)

Def of LU: If $\mathbf{A}_{n \times n}$ is invertible, then $\exists \mathbf{L}, \mathbf{U}$: 单位下三角, $\exists \mathbf{U}$: 可逆上三角. s.t. $\mathbf{A} = \mathbf{LU}$ (**unique**)

LU Factorization: By Gaussian elimination, $\mathbf{A}_{n \times n} = (\prod_{k=1}^{n-1} \mathbf{L}_k)^{-1} \mathbf{U} =: \mathbf{LU}$ $\mathbf{L}_k$: Lower triangular with 1s on the diagonal (by row operation on single column)

1. 如果 $\mathbf{L}$ 是单位下三角矩阵, 且只有第 k 列有非零元素在对角线下, 则 $\mathbf{L}^{-1}$ 矩阵是 $\mathbf{L}$ 对角线不动, 其余为相反数.

2. 如果 $\mathbf{L}_1, \mathbf{L}_2$ 矩阵是单位下三角矩阵, 且 $\mathbf{L}_1 \mathbf{L}_2$ 仅在第 $1 \sim k | k+1 \sim n$ 列有非零元素在对角线下, 则 $\mathbf{L}_1 \mathbf{L}_2$ 矩阵是他们的"拼接".

3. By LU factorization, $\mathbf{A} = \mathbf{LU} \Rightarrow$ By solving $\mathbf{Ly} = \mathbf{b}$ and $\mathbf{Ux} = \mathbf{y}$, we can solve SLE $\mathbf{Ax} = \mathbf{b}$. 用 **FS** 和 **BS** 算法

Error Analysis of GE: Gaussian elimination is an **unstable** algorithm. (i.e. floating point arithmetic can have a disastrous effect on the computed solution.)

2.3 LUPP | LUCP | GEPP | GECP

Permutation Matrix: For a permutation π , $\mathbf{P}_\pi := [\mathbf{e}_{\pi(1)}, \dots, \mathbf{e}_{\pi(n)}]$ ps: 写的时候是对 i 第 i 行的 1 移动到第 $\pi(i)$ 行. 注意: 用另一方式定义的 $\mathbf{P}_\pi$ 会让后面面对 π 结论不一样

· 左边乘 $\mathbf{P}_\pi$: 交换行 ($\mathbf{P}_\pi$ 的行 代表具体行的交换方法, π 也代表交换行的方法)

· 右边乘 $\mathbf{P}_\pi$: 交换列 ($\mathbf{P}_\pi$ 的列 代表具体列的交换方法, π^{-1} 才代表交换列的方法)

· Proposition: $\mathbf{P}_\pi$ is orthogonal, i.e. $\mathbf{P}_\pi^\top = \mathbf{P}_\pi^{-1}$ · 对于二阶置换矩阵 (**Involution**): $\mathbf{P}_\pi^2 = \mathbf{I}$, i.e. $\mathbf{P}_\pi^{-1} = \mathbf{P}_\pi$ · **GEPP|GECP:** 交换行 (行和列) 使每次 $|u_{kk}|$ 最大 (**stable**)

Backward Stable of GEPP: $\forall$ SLE sol $\hat{\mathbf{x}}$ by GEPP, $\exists \Delta \mathbf{A}$ s.t. $(\mathbf{A} + \Delta \mathbf{A})\hat{\mathbf{x}} = \mathbf{b}$ and $\frac{\|\Delta \mathbf{A}\|_\infty}{\|\mathbf{A}\|_\infty} \leq p(n) 2^{n-1} \varepsilon_m$ $p(n)$ is a cubic polynomial

2.4 QR Factorization Method

QR Factorization: QR factorization of $\mathbf{A}_{n \times n}$ is $\mathbf{A} = \mathbf{QR}$, where $\mathbf{Q}$ is orthogonal (i.e. $\mathbf{Q}^T = \mathbf{Q}^{-1}$) and $\mathbf{R}$ is non-singular upper triangular.

· Def of $\mathbf{Q}$ is orthogonal is equivalent to $\mathbf{Q}^T \mathbf{Q} = \mathbf{I}$. (i.e. columns of $\mathbf{Q}$ forming an orthonormal set.) · QR produce **large errors**

· **Orthogonal Projection:** $\text{proj}_{\mathbf{u}}(\mathbf{x}) = \frac{\mathbf{x} \cdot \mathbf{u}}{\|\mathbf{u}\|_2^2} \mathbf{u} = (\mathbf{u}^\top \mathbf{x}) \mathbf{u}$ if $\|\mathbf{u}\|_2 = 1$. $\mathbf{x} = \text{proj}_{\mathbf{u}}(\mathbf{x})$ parallel to $\mathbf{u}$ + $(\mathbf{x} - \text{proj}_{\mathbf{u}}(\mathbf{x}))$ orthogonal to $\mathbf{u}$

Error Analysis of MGS: The matrices $\hat{\mathbf{Q}}$ and $\hat{\mathbf{Q}}$ computed by MGS satisfies: $\hat{\mathbf{Q}}^\top \hat{\mathbf{Q}} = \mathbf{I} + \Delta \mathbf{I}$ $\hat{\mathbf{Q}} \hat{\mathbf{R}} = \mathbf{A} + \Delta \mathbf{A}$ where:

$$\|\Delta \mathbf{I}\|_2 \leq \frac{\alpha_1(n) \varepsilon_m \kappa_2(\mathbf{A})}{1 - \alpha_1(n) \varepsilon_m \kappa_2(\mathbf{A})}, \text{ if } 1 - \alpha_1(n) \varepsilon_m \kappa_2(\mathbf{A}) > 0 \quad \|\Delta \mathbf{A}\|_2 \leq \alpha_2(n) \varepsilon_m \|\mathbf{A}\|_2 \text{ ps: } \alpha_1(n), \alpha_2(n) \text{ 只由 } n \text{ 决定}$$

2.5 Householder QR factorisation

Householder QR: (Most) **stable** algorithm **Householder Reflection Matrix:** $\mathbf{H}(\mathbf{w}) := \mathbf{I} - 2 \frac{\mathbf{w} \mathbf{w}^\top}{\|\mathbf{w}\|_2^2}$ $\mathbf{H}(\mathbf{w}) \mathbf{x} = \mathbf{x} - 2 \text{proj}_{\mathbf{w}}(\mathbf{x})$

· $\mathbf{H}(\mathbf{w})$ is **orthogonal** and **symmetric** (In the Algorithm is $\mathbf{Q}_k, \mathbf{H}_k$) · $\mathbf{H}(\mathbf{w}) \mathbf{x}$ 作用为让 $\mathbf{x}$ 向量对 $\mathbf{w}$ 的法向量 (超平面) 上进行反射

Error Analysis of Householder QR: Let $\hat{\mathbf{x}}$ denote sol of $\mathbf{Ax} = \mathbf{b}$ by Householder QR+SQR. Then $(\mathbf{A} + \Delta \mathbf{A})\hat{\mathbf{x}} = \mathbf{b}$, where $\|\Delta \mathbf{A}\|_2 \leq \alpha n^{\frac{5}{2}} \varepsilon_m \|\mathbf{A}\|_2$ α small constant

3 Iterative Methods for SLE

Iterative Method: For SLE $\mathbf{Ax} = \mathbf{b}$, if $\mathbf{A} = \mathbf{M} + \mathbf{N}$ with $\mathbf{M}$ is *simpler structure* than $\mathbf{A}$. (例如: 对角/三角/对称矩阵) $\Rightarrow$ 迭代求近似解 Diff choices $\mathbf{M}, \mathbf{N} \rightarrow$ different iterative methods.

Error Analysis: Error $\mathbf{e}_k := \mathbf{x} - \mathbf{x}_k \Rightarrow \mathbf{e}_k = \mathbf{R}^k \mathbf{e}_0$ $\mathbf{R} := -\mathbf{M}^{-1} \mathbf{N}$ **Theorem:** If $\|\mathbf{R}\|_p < 1$, then $\lim_{k \rightarrow \infty} \|\mathbf{e}_k\|_p = 0 \Leftrightarrow \lim_{k \rightarrow \infty} \mathbf{x}_k = \mathbf{x}$ (Convergence) **GAU** 收敛快于 **JAC**

JAC|GAU: If $\mathbf{A}$ is **strictly diagonally dominant** [i.e. $|a_{ii}| > \sum_{j \neq i} |a_{ij}| \forall i$ (对角线元素大于该行其他元素之和)], then converges for $p = \infty$. For $p = 1$ 看列和/转置.

4 Algorithms

4.1 DP MV MM

DP Dot product	$C = \sum_{i=1}^n 2 = 2n = \mathcal{O}(n)$
Input: $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$	
Output: $s = \mathbf{x} \cdot \mathbf{y} = \mathbf{x}^\top \mathbf{y}$	
1: $s = 0$	
2: for $i = 1, \dots, n$ do	
3: $s = s + x_i y_i$	
4: end for	

4.2 LU FS BS | GE GEPP GECP | LUPP LUCP

LU LU factorisation
$C(n) = \sum_{k=1}^{n-1} (n-k)(1+2(n-k+1)) = \frac{2}{3}n^3 + \frac{1}{2}n^2 - \frac{7}{6}n$
Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$
Output: $\mathbf{L}, \mathbf{U} \in \mathbb{R}^{n \times n}$ s.t. $\mathbf{A} = \mathbf{LU}$
1: $\mathbf{L} = \mathbf{I}, \mathbf{U} = \mathbf{A}$
2: for $k = 1, \dots, n-1$ do
3: for $j = k+1, \dots, n$ do
4: $l_{jk} = u_{jk}/u_{kk}$
5: $(u_{jk}, \dots, u_{jn}) = (u_{jk}, \dots, u_{jn}) - l_{jk}(u_{kk}, \dots, u_{kn})$
6: end for
7: end for

GE Gaussian elimination
$C(n) = \frac{2}{3}n^3 + \frac{5}{2}n^2 - \frac{7}{6}n = \mathcal{O}(n^3)$
Input: $\mathbf{A} \in \mathbb{R}^{n \times n}, \mathbf{b} \in \mathbb{R}^n$
Output: $\mathbf{x} \in \mathbb{R}^n$ s.t. $\mathbf{Ax} = \mathbf{b}$
1: Find LU factorisation of $\mathbf{A}$ # LU Algorithm
2: Solve $\mathbf{Ly} = \mathbf{b}$ using forward substitution. # FS Algorithm
3: Solve $\mathbf{Ux} = \mathbf{y}$ using back substitution. # BS Algorithm

LUPP LU factorisation with partial pivoting (not unique)
$C(n) = \frac{2}{3}n^3 + n^2 - \frac{5}{3}n$

Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$
Output: $\mathbf{L}, \mathbf{U}, \mathbf{P} \in \mathbb{R}^{n \times n}$ s.t. $\mathbf{PA} = \mathbf{LU}$ with $\mathbf{L}$ 单位下三角, $\mathbf{U}$ 可逆上三角
1: $\mathbf{U} = \mathbf{A}, \mathbf{L} = \mathbf{I}, \mathbf{P} = \mathbf{I}$
2: for $k = 1, \dots, n-1$ do
3: Choose $i \in \{k, \dots, n\}$ which maximizes $ u_{ik} $ # Max Algorithm
4: Exchange row $(u_{kk}, \dots, u_{kn})$ with $(u_{ik}, \dots, u_{in})$ # 第 k 行和第 i 行的 $k \sim n$ 列交换
5: Exchange row $(l_{k1}, \dots, l_{k,k-1})$ with $(l_{i1}, \dots, l_{i,k-1})$ # 第 k 行和第 i 行的 $1 \sim k-1$ 列交换
6: Exchange row $(p_{k1}, \dots, p_{kn})$ with $(p_{i1}, \dots, p_{in})$ # 第 k 行和第 i 行的全部列交换
7: for $j = k+1, \dots, n$ do
8: $l_{jk} = u_{jk}/u_{kk}$
9: $(u_{jk}, \dots, u_{jn}) = (u_{jk}, \dots, u_{jn}) - l_{jk}(u_{kk}, \dots, u_{kn})$
10: end for
11: end for

4.3 (M)GS SQR Householder-QR

SQR (SLE) via QR-Factorization	$C(n) = 2n^3 + 4n^2 + n$
Input: $\mathbf{A} \in \mathbb{R}^{n \times n}, \mathbf{b} \in \mathbb{R}^n$	
Output: $\mathbf{x} \in \mathbb{R}^n$ with $\mathbf{Ax} = \mathbf{b}$	
1: Find QR factorization of $\mathbf{A}$ # GS MGs Algorithm	
2: Solve $\mathbf{Qy} = \mathbf{b}$ # MV Algorithm # ps: $\mathbf{y} = \mathbf{Q}^{-1}\mathbf{b} = \mathbf{Q}^\top \mathbf{b}$	
3: Solve $\mathbf{Rx} = \mathbf{y}$ using Algorithm BS # BS Algorithm	

Sum Notation: $\sum_{k=1}^n k = \frac{n(n+1)}{2} = \frac{1}{2}n^2 + \frac{1}{2}n$
$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} = \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n$
$\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2 = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2$
Rank Theorem: $\mathbf{A}_{m \times n}, m \geq n: [\mathbf{Ax} = \mathbf{0} \text{ iff } \mathbf{x} = \mathbf{0}] \Leftrightarrow [\text{rank}(\mathbf{A}) = n]$
Cauchy-Schwarz Inequality: For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \mathbf{x} \cdot \mathbf{y} = \mathbf{x}^\top \mathbf{y} \leq \ \mathbf{x}\ _2 \ \mathbf{y}\ _2$
Determinant: $\det(\mathbf{AB}) = \det(\mathbf{A})\det(\mathbf{B})$ Orthogonal: If $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$
Eigen: $\mathbf{A}\mathbf{v} = \lambda \mathbf{v} \quad \xi_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I}) = \prod_i (\lambda_i - \lambda)$

4.4 Iterative Method | OP MAX

GI JAC GAU	$\text{JAC}: C = \mathcal{O}(k_{\max} n^2) \quad \quad \text{GAU}: C = \mathcal{O}(k_{\max} n^2)$
Input: $\mathbf{A} = \mathbf{M} + \mathbf{N} \in \mathbb{R}^{n \times n}, \mathbf{b} \in \mathbb{R}^n, \mathbf{x}_0 \in \mathbb{R}^n, \epsilon_r > 0$	
Output: $\mathbf{x}_k \in \mathbb{R}^n$ with $\mathbf{Ax}_k \approx \mathbf{b}$	
1: $k = 0$ # JAC: $C = 2n^2/\text{it}$ GAU: $C = 2n^2/\text{it}$	
2: while $\ \mathbf{Ax}_k - \mathbf{b}\ _p > \epsilon_r$ do # e.g. $\epsilon_r = 10^{-3} \ \mathbf{b}\ _p$	
3: $k = k + 1$ # 上面给的 C 不含这步	
4: Compute $\mathbf{y}_k = \mathbf{b} - \mathbf{Nx}_{k-1}$	
5: Solve $\mathbf{Mx}_k = \mathbf{y}_k$ # $\downarrow \mathbf{A} = \mathbf{D} + \mathbf{L} + \mathbf{U}$	
6: end while # JAC: $\mathbf{M} = \mathbf{D}, \mathbf{N} = \mathbf{L} + \mathbf{U}; \text{GAU: } \mathbf{M} = \mathbf{D} + \mathbf{L}, \mathbf{N} = \mathbf{U}$	

MV Matrix-vector multiplication
$C = \sum_{i=1}^m 2n = 2nm = \mathcal{O}(mn)$
Input: $\mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{x} \in \mathbb{R}^n$
Output: $\mathbf{y} = \mathbf{Ax}$
1: for $i = 1, \dots, m$ do
2: $y_i = \mathbf{a}_i^\top \mathbf{x}$ # DP Algorithm
3: end for

FS Forward substitution
$C(n) = \sum_{j=1}^n [2(j-1) + 1] = n^2 = \mathcal{O}(n^2)$
Input: $\mathbf{L} \in \mathbb{R}^{n \times n}$ (lower triangular), $\mathbf{b} \in \mathbb{R}^n$
Output: $\mathbf{y} \in \mathbb{R}^n$ such that $\mathbf{Ly} = \mathbf{b}$
1: for $j = 1, \dots, n$ do
2: $y_j = \frac{1}{l_{jj}} \left(b_j - \sum_{k=1}^{j-1} l_{jk} y_k \right)$
3: end for

GEPP Gaussian elimination with partial pivoting
$C(n) = \frac{2}{3}n^3 + 3n^2 - \frac{5}{3}n = \mathcal{O}(n^3)$
Input: $\mathbf{A} \in \mathbb{R}^{n \times n}, \mathbf{b} \in \mathbb{R}^n$ # numpy.linalg.solve use it
Output: $\mathbf{x} \in \mathbb{R}^n$ satisfying $\mathbf{Ax} = \mathbf{b}$
1: Find $\mathbf{PA} = \mathbf{LU}$ using Algorithm LUPP # LUPP Algorithm
2: Solve $\mathbf{Ly} = \mathbf{Pb}$ by forward substitution # FS Algorithm
3: Solve $\mathbf{Ux} = \mathbf{y}$ by back substitution # BS Algorithm

LUCP LU factorisation with complete pivoting (not unique)

$C(n) = n^3 + n^2 - 2n$
Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$
Output: $\mathbf{L}, \mathbf{U}, \mathbf{P}, \mathbf{Q} \in \mathbb{R}^{n \times n}$ s.t. $\mathbf{PAQ} = \mathbf{LU}$ with $\mathbf{L}$ 单位下三角, $\mathbf{U}$ 可逆上三角
1: $\mathbf{U} = \mathbf{A}, \mathbf{L} = \mathbf{I}, \mathbf{P} = \mathbf{I}, \mathbf{Q} = \mathbf{I}$
2: for $k = 1, \dots, n-1$ do
3: Choose $(i, j) \in \{(k, k), \dots, (n, n)\}$ which maximizes $ u_{ij} $ # Max Algorithm
4: Exchange row $(u_{kk}, \dots, u_{kn})$ with $(u_{ik}, \dots, u_{in})$ # 第 k 行和第 i 行的 $k \sim n$ 列交换 U
5: Exchange row $(l_{k1}, \dots, l_{k,k-1})$ with $(l_{i1}, \dots, l_{i,k-1})$ # 第 k 行和第 i 行的 $1 \sim k-1$ 列交换 L
6: Exchange row $(p_{k1}, \dots, p_{kn})$ with $(p_{i1}, \dots, p_{in})$ # 第 k 行和第 i 行的全部列交换 P
7: Exchange column $(u_{1k}, \dots, u_{nk})$ with $(u_{1j}, \dots, u_{nj})$ # 第 k 列和第 j 列的全部行交换 U
8: Exchange column $(q_{1k}, \dots, q_{nk})$ with $(q_{1j}, \dots, q_{nj})$ # 第 k 列和第 j 列的全部行交换 Q
9: for $m = k+1, \dots, n$ do
10: $l_{mk} = u_{mk}/u_{kk}$
11: $(u_{mk}, \dots, u_{mn}) = (u_{mk}, \dots, u_{mn}) - l_{mk}(u_{kk}, \dots, u_{kn})$
12: end for
13: end for

(M)GS (Modified) Gram-Schmidt	$C(n) = 2n^3 + n^2 + n$
Input: $\mathbf{A} \in \mathbb{R}^{n \times n}$, with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$	
Output: $\mathbf{Q} \in \mathbb{R}^{n \times n}$, with orthonormal columns $\mathbf{q}_1, \dots, \mathbf{q}_n$	
$\mathbf{R} \in \mathbb{R}^{n \times n}$ non-singular upper triangular matrix; s.t. $\mathbf{A} = \mathbf{QR}$	
1: $\mathbf{R} = \mathbf{0}$	
2: for $j = 1, \dots, n$ do	
3: $\mathbf{q}_j = \mathbf{a}_j$	
4: for $k = 1, \dots, j-1$ do	
5: $r_{kj} = \mathbf{q}_k^\top \mathbf{a}_j$ $(r_{kj} = \mathbf{q}_k^\top \mathbf{q}_j)$	
6: $\mathbf{q}_j = \mathbf{q}_j - r_{kj} \mathbf{q}_k$	
7: end for	
8: $r_{jj} = \ \mathbf{q}_j\ _2 = \sqrt{\mathbf{q}_j^\top \mathbf{q}_j}$	
9: $\mathbf{q}_j = \frac{\mathbf{q}_j}{r_{jj}}$	
10: end for	

OP Outer product	$C = \sum_{i=1}^n \sum_{j=1}^n 1 = n^2 = \mathcal{O}(n^2)$
Input: $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$	
Output: $\mathbf{A} = \mathbf{xy}^\top$	
1: for $i = 1, \dots, n$ do	
2: for $j = 1, \dots, n$ do	
3: $a_{ij} = x_i y_j$	
4: end for	
5: end for	

MM Matrix-matrix multiplication
$C = \sum_{i=1}^m \sum_{j=1}^p 2n = 2mnp = \mathcal{O}(mnp)$
Input: $\mathbf{A} \in \mathbb{R}^{m \times n}, \mathbf{B} \in \mathbb{R}^{n \times p}$
Output: $\mathbf{C} = \mathbf{AB}$
1: for $i = 1, \dots, m$ do
2: for $j = 1, \dots, p$ do
3: $c_{ij} = \mathbf{a}_i^\top \mathbf{b}_j$ # DP Algorithm
4: end for
5: end for

BS Back substitution
$C(n) = \sum_{j=1}^n [2(n-j) + 1] = n^2 = \mathcal{O}(n^2)$
Input: $\mathbf{U} \in \mathbb{R}^{n \times n}$ (upper triangular), $\mathbf{y} \in \mathbb{R}^n$
Output: $\mathbf{x} \in \mathbb{R}^n$ such that $\mathbf{Ux} = \mathbf{y}$
1: for $j = n, \dots, 1$ do
2: $x_j = \frac{1}{u_{jj}} \left(y_j - \sum_{k=j+1}^n u_{jk} x_k \right)$
3: end for

GECP Gaussian elimination with complete pivoting
$C(n) = n^3 + 3n^2 - 2n = \mathcal{O}(n^3)$
Input: $\mathbf{A} \in \mathbb{R}^{n \times n}, \mathbf{b} \in \mathbb{R}^n$
Output: $\mathbf{x} \in \mathbb{R}^n$ satisfying $\mathbf{Ax} = \mathbf{b}$
1: Find $\mathbf{PAQ} = \mathbf{LU}$ using Algorithm LUCP # LUCP Algorithm
2: Solve $\mathbf{Ly} = \mathbf{Pb}$ by forward substitution # FS Algorithm
3: Solve $\mathbf{Uz} = \mathbf{y}$ by back substitution # BS Algorithm
4: Compute $\mathbf{x} = \mathbf{Qz}$

5 Appendix

5.1 Useful Theorems/Formulas

Rank Theorem: These statements are equivalent: $[\mathbf{Ax} = \mathbf{0} \text{ iff } \mathbf{x} = \mathbf{0}] \Leftrightarrow [\text{columns of } \mathbf{A} \text{ are linearly independent}] \Leftrightarrow [\mathbf{A} \text{ is invertible}] \Leftrightarrow [\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top) = n]$

Diagonalisable: $\mathbf{A}$ is diagonalisable iff have n LI eigenvectors. (e.g. symmetric matrix - Spectral Theorem) | Always has Jordan Form.

Spectral Theorem: If $\mathbf{A}$ is symmetric, then $\exists$ orthonormal $\mathbf{Q}$ s.t. $\mathbf{A} = \mathbf{Q}^\top \mathbf{\Lambda} \mathbf{Q}$ (Orthogonally diagonalisable) $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$

5.2 Others

Where the error come from calucation:

1. If your calculation involves 15 or more significant digits, you will likely encounter rounding errors. (Since $\epsilon_m \approx 10^{-16}$ (relative accuracy), which is about 15 significant digits.)
2. **For GE Algorithm:** It is an unstable algorithm. e.g. $\mathbf{A} = [10^{-20} \ 1 \ \dots \ 1 \ 1] \Rightarrow$ By GE Algorithm, $\tilde{\mathbf{L}}\hat{\mathbf{U}} = \mathbf{A} + [0 \ 0 \ \dots \ 0 \ -1]$ (error)
3. **MGS vs. GS:** In floating point arithmetic, the computed vectors $\{\hat{q}_1, \dots, \hat{q}_{j-1}\}$ will not be orthonormal. Algorithm MGS takes this into account when computing $\hat{q}_j$; Algorithm GS does not.